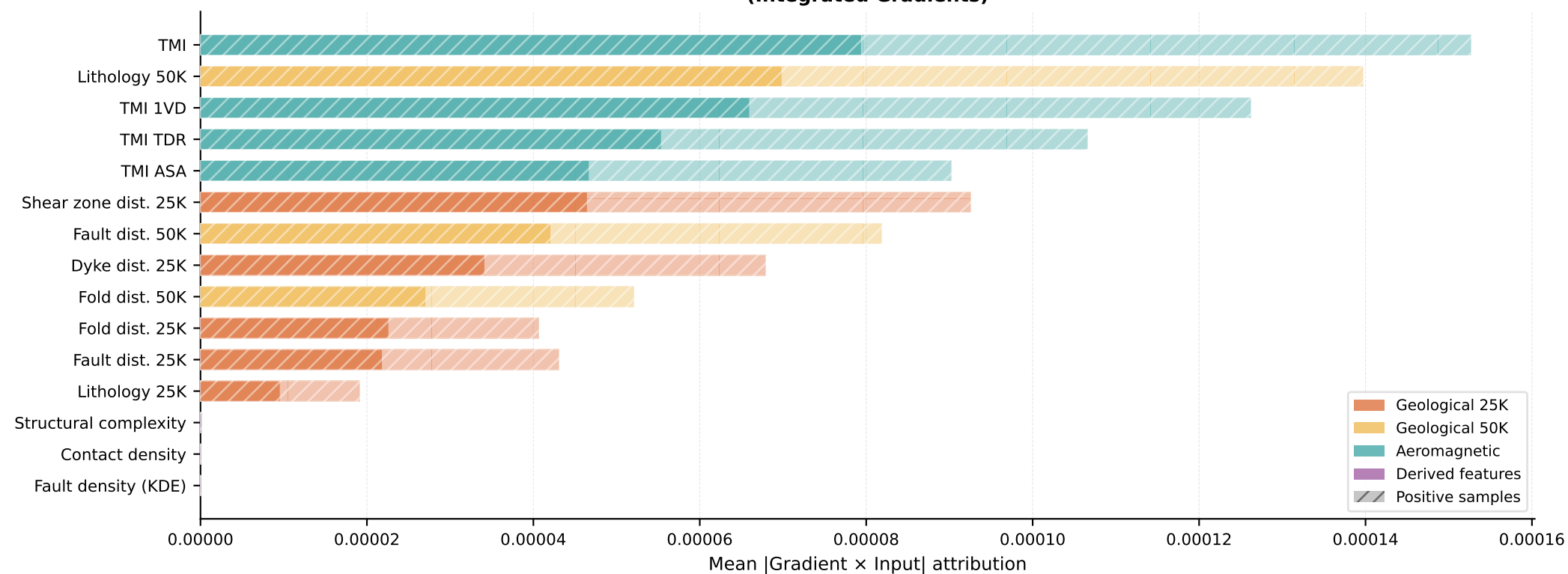
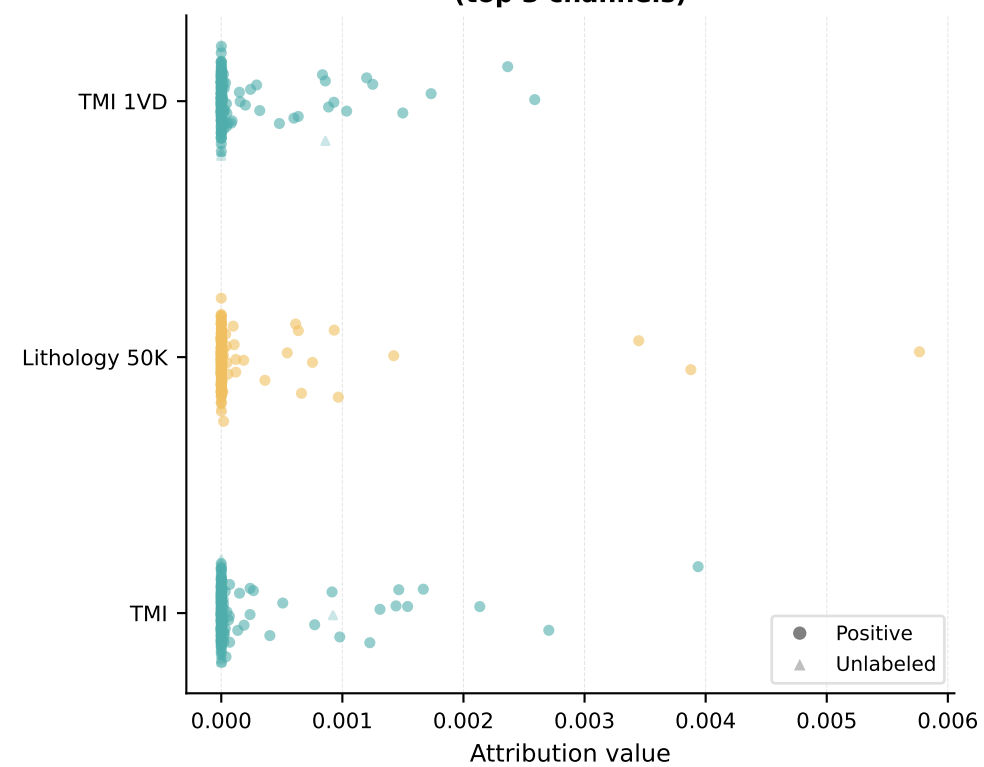


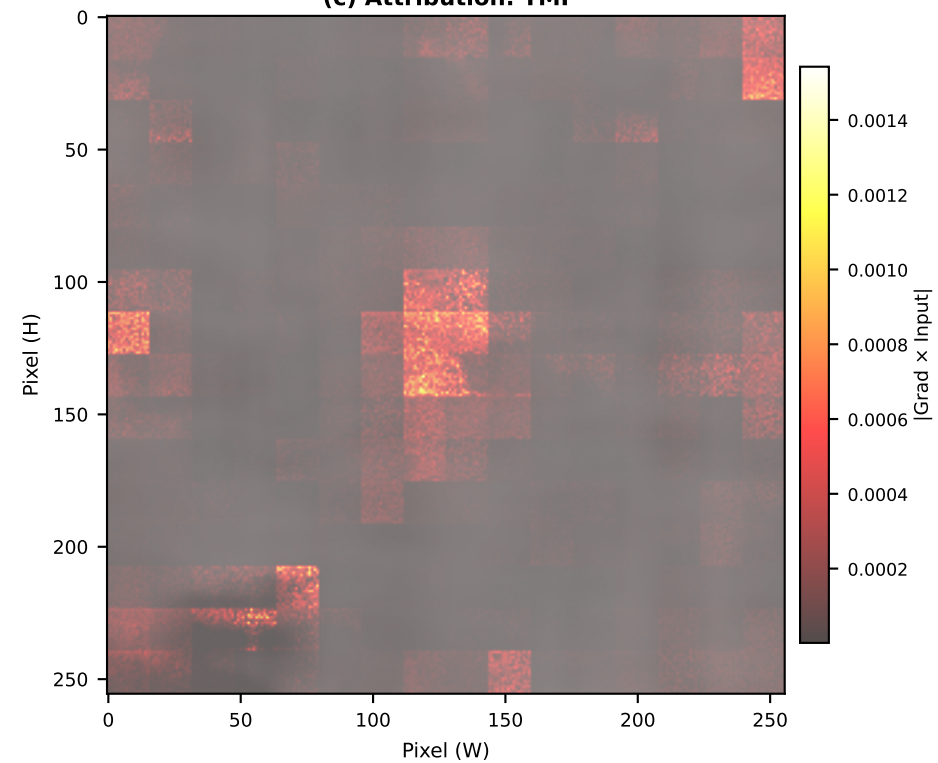
**Figure 7 — Explainability Analysis: Channel Feature Importance**  
**Method: Integrated Gradients | n=300 samples (fold 0) | Depth: 0-500 m**  
**(a) Channel feature importance (Integrated Gradients)**



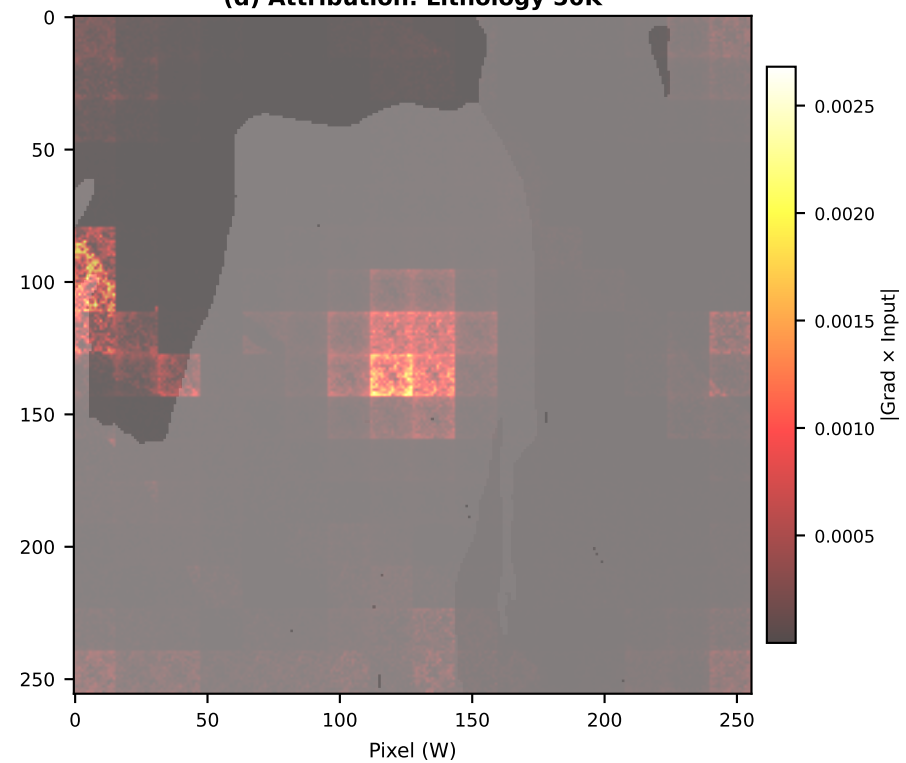
**(b) Per-sample attribution (top 3 channels)**



**(c) Attribution: TMI**



**(d) Attribution: Lithology 50K**



**(e) Attribution: TMI 1VD**

